DocCopilot AI

Mathematical Foundations for Document-Level
LLM Evaluation and Processing

Lecture Notes Style Technical Report

Technical Documentation v2.0

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Abstract

This document provides rigorous mathematical foundations for the DocCopilot AI framework. Each section corresponds to a demo module and includes formal definitions, step-by-step derivations, and detailed symbol explanations. The goal is clarity through mathematics: every formula is accompanied by precise definitions of its components.

Contents

1 Demo 1: Evaluation Metrics **3**
1.1 N-grams: The Building Block . . . 3
1.2 ROUGE: Recall-Oriented Understudy for Gisting Evaluation . . . 3
1.2.1 ROUGE-N: N-gram Recall . . . 3
1.2.2 ROUGE-L: Longest Common Subsequence . . . 4
1.3 BLEU: Bilingual Evaluation Understudy . . . 5
1.4 BERTScore: Contextual Similarity . . . 6

2 Demo 2: A/B Testing Framework **6**
2.1 Hypothesis Testing Framework . . . 6
2.2 Welch's t-Test . . . 7
2.3 Effect Size: Cohen's d . . . 8
2.4 Confidence Interval . . . 8
2.5 Sample Size Calculation . . . 8

3 Demo 3: Document Coherence Evaluation **9**
3.1 Lexical Cohesion . . . 9
3.2 Semantic Coherence via Word Overlap . . . 9
3.3 Overall Coherence Score . . . 10
3.4 Why BLEU Fails for Discourse . . . 11

4 Demo 4: Attention Complexity Analysis **11**
4.1 Self-Attention Mechanism . . . 11
4.2 Full Concatenation Complexity . . . 12
4.3 Hierarchical Attention Complexity . . . 12
4.4 Speedup Analysis . . . 13
4.5 Memory Analysis . . . 13

5 Demo 5: LLM Document Processing **13**
5.1 Autoregressive Language Modeling . . . 13
5.2 Summarization . . . 14
5.3 Question Answering . . . 14

6 Demo 6: LLM-as-Judge Evaluation **14**
6.1 Multi-Dimensional Scoring . . . 14
6.2 Pairwise Comparison . . . 15
6.3 Tournament Ranking . . . 15

7 Demo 7: API Architecture **16**
7.1 REST API Design . . . 16

8 Advanced Topics **16**
8.1 Hierarchical Context Modeling . . . 16
8.2 Contrastive Evaluation . . . 17
8.3 Zero-Resource Domain Adaptation . . . 17
8.4 Oracle Experiments . . . 17

9 Summary of Key Formulas **18**

1 Demo 1: Evaluation Metrics

Notation 1.1 (Basic Setup)**.** Throughout this section:

$$Y = (y_1, y_2, \ldots, y_n) \quad \text{Reference text with } n \text{ tokens} \tag{1}$$
$$\hat{Y} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_m) \quad \text{Generated text with } m \text{ tokens} \tag{2}$$
$$\mathcal{V} = \{w_1, w_2, \ldots, w_{|\mathcal{V}|}\} \quad \text{Vocabulary of all possible tokens} \tag{3}$$

1.1 N-grams: The Building Block

Definition 1.1 (N-gram)**.** An **n-gram** is a contiguous sequence of n tokens from a text. For sequence $S = (s_1, s_2, \ldots, s_k)$, the set of all n-grams is:

$$\boxed{\text{ngrams}_n(S) = \{(s_i, s_{i+1}, \ldots, s_{i+n-1}) \,:\, i \in \{1, 2, \ldots, k-n+1\}\}} \tag{4}$$

Symbol explanation:

- s_i = the i-th token in sequence S
- n = the length of each n-gram (e.g., $n = 1$ for unigrams, $n = 2$ for bigrams)
- $k - n + 1$ = total number of n-grams extractable from a sequence of length k

Example 1.1. For $S =$ ("the","quick","brown","fox"):

$$\text{ngrams}_1(S) = \{(\text{the}), (\text{quick}), (\text{brown}), (\text{fox})\} \quad \text{4 unigrams} \tag{5}$$
$$\text{ngrams}_2(S) = \{(\text{the, quick}), (\text{quick, brown}), (\text{brown, fox})\} \quad \text{3 bigrams} \tag{6}$$

Definition 1.2 (Count Function)**.** $\text{count}_g(S)$ returns the number of times n-gram g appears in sequence S:

$$\text{count}_g(S) = \sum_{i=1}^{|S|-n+1} \mathbf{1}\left[(s_i, \ldots, s_{i+n-1}) = g\right] \tag{7}$$

where $\mathbf{1}[\cdot]$ is the indicator function:

$$\mathbf{1}[P] = \begin{cases} 1 & \text{if proposition } P \text{ is true} \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

1.2 ROUGE: Recall-Oriented Understudy for Gisting Evaluation

ROUGE measures how much of the **reference** content appears in the generated text.

1.2.1 ROUGE-N: N-gram Recall

Definition 1.3 (ROUGE-N Recall)**.**

$$\boxed{\text{ROUGE-N}_{\text{recall}} = \frac{\displaystyle\sum_{g \in \text{ngrams}_n(Y)} \min\big(\text{count}_g(\hat{Y}), \text{count}_g(Y)\big)}{\displaystyle\sum_{g \in \text{ngrams}_n(Y)} \text{count}_g(Y)}} \tag{9}$$

Symbol explanation:

- Numerator: Count of n-grams from reference Y that also appear in generated $\hat{Y}$
- $\min(\cdot, \cdot)$: Prevents counting the same n-gram more times than it appears in reference
- Denominator: Total count of n-grams in the reference (normalization factor)

Definition 1.4 (ROUGE-N Precision)**.**

$$\text{ROUGE-N}_{\text{precision}} = \frac{\sum_{g \in \text{ngrams}_n(\hat{Y})} \min\big(\text{count}_g(\hat{Y}), \text{count}_g(Y)\big)}{\sum_{g \in \text{ngrams}_n(\hat{Y})} \text{count}_g(\hat{Y})} \tag{10}$$

Difference from recall: Iterates over n-grams in $\hat{Y}$ (generated) and normalizes by $|\hat{Y}|$.

Definition 1.5 (ROUGE-N F_1 Score)**.** The harmonic mean of precision and recall:

$$\boxed{\text{ROUGE-N}_{F_1} = \frac{2 \cdot P \cdot R}{P + R} = \frac{2}{\frac{1}{P} + \frac{1}{R}}} \tag{11}$$

where $P = \text{ROUGE-N}_{\text{precision}}$ and $R = \text{ROUGE-N}_{\text{recall}}$.

Remark 1.1 (Why Harmonic Mean?)**.** The harmonic mean penalizes extreme imbalance. If $P = 1.0$ but $R = 0.01$:

$$F_1 = \frac{2 \cdot 1.0 \cdot 0.01}{1.0 + 0.01} = \frac{0.02}{1.01} \approx 0.02 \tag{12}$$

Arithmetic mean would give $(1.0 + 0.01)/2 = 0.505$, which is misleading.

1.2.2 ROUGE-L: Longest Common Subsequence

Definition 1.6 (Subsequence vs. Substring)**.** A **subsequence** of S is any sequence that can be derived by deleting some (or no) elements without changing the order. A **substring** must be contiguous.

$$S = (A, B, C, D, E) \tag{13}$$

$$(A, C, E) \text{ is a subsequence (non-contiguous OK)} \tag{14}$$

$$(B, C, D) \text{ is a substring (contiguous)} \tag{15}$$

Definition 1.7 (Longest Common Subsequence (LCS))**.** For sequences $X = (x_1, \ldots, x_m)$ and $Y = (y_1, \ldots, y_n)$, define $L(i, j)$ as the length of LCS of prefixes $X[1:i]$ and $Y[1:j]$:

$$\boxed{L(i,j) = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ L(i-1, j-1) + 1 & \text{if } x_i = y_j \\ \max\big(L(i-1, j), L(i, j-1)\big) & \text{if } x_i \neq y_j \end{cases}} \tag{16}$$

Intuition:

- If last characters match: extend LCS by 1
- If they don't match: take the better of excluding x_i or excluding y_j

Definition 1.8 (ROUGE-L Scores)**.** Let $\ell = L(m, n) = |\text{LCS}(\hat{Y}, Y)|$. Then:

$$R_{\text{LCS}} = \frac{\ell}{n} \qquad \text{(Recall: fraction of reference covered)} \tag{17}$$

$$P_{\text{LCS}} = \frac{\ell}{m} \qquad \text{(Precision: fraction of generated that matches)} \tag{18}$$

$$F_{\text{LCS}} = \frac{(1 + \beta^2) \cdot P_{\text{LCS}} \cdot R_{\text{LCS}}}{R_{\text{LCS}} + \beta^2 \cdot P_{\text{LCS}}} \tag{19}$$

Parameter β: Controls precision-recall tradeoff.

- $\beta = 1$: Equal weight (standard F_1)
- $\beta > 1$: Favors recall
- $\beta < 1$: Favors precision

1.3 BLEU: Bilingual Evaluation Understudy

BLEU is **precision-focused** with a penalty for short outputs.

Definition 1.9 (Modified Precision)**.** To prevent gaming by repeating common words:

$$\boxed{p_n = \frac{\sum_{\hat{Y}} \sum_{g \in \text{ngrams}_n(\hat{Y})} \min\Big(\text{count}_g(\hat{Y}), \max_{Y \in \mathcal{R}} \text{count}_g(Y)\Big)}{\sum_{\hat{Y}} \sum_{g \in \text{ngrams}_n(\hat{Y})} \text{count}_g(\hat{Y})}} \tag{20}$$

Key insight: The max over references $\mathcal{R}$ handles multiple valid references.
The min **clips:** If generated has "the the the" but reference has one "the", count only 1.

Definition 1.10 (Brevity Penalty (BP))**.** Let $c = |\hat{Y}|$ (generated length) and r = closest reference length:

$$\boxed{\text{BP} = \begin{cases} 1 & \text{if } c > r \quad \text{(no penalty if generated is longer)} \\ e^{1-r/c} & \text{if } c \leq r \quad \text{(exponential penalty for short outputs)} \end{cases}} \tag{21}$$

Derivation of penalty form:

- When $c = r$: $\text{BP} = e^{1-1} = e^0 = 1$ (no penalty)
- When $c = r/2$: $\text{BP} = e^{1-2} = e^{-1} \approx 0.37$ (63% penalty)
- When $c \to 0$: $\text{BP} \to 0$ (complete penalty)

Definition 1.11 (BLEU Score)**.**

$$\boxed{\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)} \tag{22}$$

Standard setting: $N = 4$, $w_n = 1/4$ (uniform weights).

Proposition 1.1 (Geometric Mean Equivalence)**.**

$$\exp\left(\sum_{n=1}^{N} w_n \log p_n\right) = \exp\left(\frac{1}{N}\sum_{n=1}^{N} \log p_n\right) = \left(\prod_{n=1}^{N} p_n\right)^{1/N} \tag{23}$$

This is the **geometric mean** of precisions p_1, p_2, p_3, p_4.

Proof. Using logarithm properties:

$$\exp\left(\frac{1}{N}\sum_{n=1}^{N} \log p_n\right) = \exp\left(\log\left(\prod_{n=1}^{N} p_n\right)^{1/N}\right) = \left(\prod_{n=1}^{N} p_n\right)^{1/N} \tag{24}$$

□

1.4 BERTScore: Contextual Similarity

Definition 1.12 (Contextual Embedding)**.** A pre-trained BERT model maps each token to a d-dimensional vector that captures context:

$$\text{BERT} : \mathcal{V}^* \to \mathbb{R}^d \tag{25}$$

For token t in context C: $\mathbf{h}_t = \text{BERT}(t \mid C) \in \mathbb{R}^d$

Definition 1.13 (Cosine Similarity)**.** For vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$:

$$\boxed{\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|} = \frac{\sum_{i=1}^d a_i b_i}{\sqrt{\sum_{i=1}^d a_i^2} \cdot \sqrt{\sum_{i=1}^d b_i^2}}} \tag{26}$$

Range: $[-1, 1]$ where 1 = identical direction, 0 = orthogonal, -1 = opposite.

Definition 1.14 (BERTScore Components)**.** Let $\hat{\mathbf{H}} = \{\hat{\mathbf{h}}_1, \ldots, \hat{\mathbf{h}}_m\}$ be embeddings of generated tokens and $\mathbf{H} = \{\mathbf{h}_1, \ldots, \mathbf{h}_n\}$ be embeddings of reference tokens.

$$\boxed{R_{\text{BERT}}} = \frac{1}{n} \sum_{j=1}^{n} \max_{i \in \{1,\ldots,m\}} \cos(\hat{\mathbf{h}}_i, \mathbf{h}_j) \qquad \text{(Recall)} \tag{27}$$

$$\boxed{P_{\text{BERT}}} = \frac{1}{m} \sum_{i=1}^{m} \max_{j \in \{1,\ldots,n\}} \cos(\hat{\mathbf{h}}_i, \mathbf{h}_j) \qquad \text{(Precision)} \tag{28}$$

$$\boxed{F_{\text{BERT}}} = 2 \cdot \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \qquad (\text{F}_1) \tag{29}$$

Intuition:

- Recall: For each reference token, find its best match in generated
- Precision: For each generated token, find its best match in reference
- Uses semantic similarity ("quick" matches "fast") unlike exact n-gram matching

2 Demo 2: A/B Testing Framework

Notation 2.1 (Experimental Setup)**.**

$$X_1, X_2, \ldots, X_{n_A} \sim F_A \qquad \text{i.i.d. samples from Model A (control)} \tag{30}$$

$$Y_1, Y_2, \ldots, Y_{n_B} \sim F_B \qquad \text{i.i.d. samples from Model B (treatment)} \tag{31}$$

$$\mu_A = \mathbb{E}[X_i], \quad \mu_B = \mathbb{E}[Y_j] \qquad \text{Population means} \tag{32}$$

$$\sigma_A^2 = \text{Var}(X_i), \quad \sigma_B^2 = \text{Var}(Y_j) \qquad \text{Population variances} \tag{33}$$

2.1 Hypothesis Testing Framework

Definition 2.1 (Statistical Hypotheses)**.**

$$H_0 : \mu_A = \mu_B \qquad \text{Null hypothesis (no difference)} \tag{34}$$

$$H_1 : \mu_A \neq \mu_B \qquad \text{Alternative hypothesis (two-tailed)} \tag{35}$$

Definition 2.2 (Sample Statistics)**. Sample Mean** (unbiased estimator of μ):

$$\boxed{\bar{X} = \frac{1}{n_A} \sum_{i=1}^{n_A} X_i} \tag{36}$$

Sample Variance (unbiased estimator of σ^2):

$$\boxed{s_A^2 = \frac{1}{n_A - 1} \sum_{i=1}^{n_A} (X_i - \bar{X})^2} \tag{37}$$

Why $n - 1$? Bessel's correction ensures $\mathbb{E}[s^2] = \sigma^2$ (unbiased).

2.2 Welch's t-Test

Definition 2.3 (Welch's t-Statistic)**.**

$$\boxed{t = \frac{\bar{Y} - \bar{X}}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}} \tag{38}$$

Components:

- Numerator: Observed difference in sample means
- Denominator: Standard error of the difference (SE)
- $\frac{s_A^2}{n_A}$: Variance of $\bar{X}$ (by CLT: $\mathrm{Var}(\bar{X}) = \sigma_A^2/n_A$)

Definition 2.4 (Standard Error)**.**

$$\mathrm{SE} = \sqrt{\mathrm{Var}(\bar{Y} - \bar{X})} = \sqrt{\mathrm{Var}(\bar{Y}) + \mathrm{Var}(\bar{X})} = \sqrt{\frac{\sigma_B^2}{n_B} + \frac{\sigma_A^2}{n_A}} \tag{39}$$

We estimate this using sample variances: $\widehat{\mathrm{SE}} = \sqrt{\frac{s_B^2}{n_B} + \frac{s_A^2}{n_A}}$

Definition 2.5 (Welch-Satterthwaite Degrees of Freedom)**.**

$$\boxed{\nu = \frac{\left(\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}\right)^2}{\frac{(s_A^2/n_A)^2}{n_A - 1} + \frac{(s_B^2/n_B)^2}{n_B - 1}}} \tag{40}$$

Derivation intuition: Approximates the distribution of a linear combination of chi-squared variables. When $\sigma_A^2 = \sigma_B^2$ and $n_A = n_B = n$, this simplifies to $\nu \approx 2n - 2$.

Theorem 2.1 (Null Distribution)**.** Under $H_0 : \mu_A = \mu_B$, the test statistic follows approximately:

$$t \sim t_\nu \quad \text{(Student's t-distribution with } \nu \text{ degrees of freedom)} \tag{41}$$

Definition 2.6 (P-value (Two-Tailed))**.**

$$\boxed{p = 2 \cdot \Pr(T > |t|) = 2 \cdot (1 - F_{t_\nu}(|t|))} \tag{42}$$

where F_{t_ν} is the CDF of the t-distribution.

Interpretation: Probability of observing a difference at least as extreme as $|t|$ if H_0 is true.

Definition 2.7 (Decision Rule)**.**

$$\text{Reject } H_0 \iff p < \alpha \tag{43}$$

where α is the significance level (typically $\alpha = 0.05$).

2.3 Effect Size: Cohen's d

Definition 2.8 (Cohen's d)**.**

$$\boxed{d = \frac{\bar{Y} - \bar{X}}{s_{\text{pooled}}}} \tag{44}$$

where the pooled standard deviation is:

$$s_{\text{pooled}} = \sqrt{\frac{s_A^2 + s_B^2}{2}} \tag{45}$$

Interpretation: Number of standard deviations separating the two means.

Proposition 2.1 (Cohen's Guidelines (1988))**.**

$\lvert d\rvert$ Range	Interpretation	Percentile Overlap
$\lvert d\rvert < 0.2$	Negligible	85% overlap
$0.2 \leq \lvert d\rvert < 0.5$	Small	67% overlap
$0.5 \leq \lvert d\rvert < 0.8$	Medium	53% overlap
$\lvert d\rvert \geq 0.8$	Large	43% overlap

2.4 Confidence Interval

Definition 2.9 ($(1-\alpha)$ Confidence Interval)**.**

$$\boxed{\text{CI}_{1-\alpha} = \left[(\bar{Y} - \bar{X}) - t_{\nu,1-\alpha/2} \cdot \text{SE}, \quad (\bar{Y} - \bar{X}) + t_{\nu,1-\alpha/2} \cdot \text{SE}\right]} \tag{46}$$

Components:

- $(\bar{Y} - \bar{X})$: Point estimate of $\mu_B - \mu_A$
- $t_{\nu,1-\alpha/2}$: Critical value (e.g., $t_{97,0.975} \approx 1.98$ for 95% CI)
- SE: Standard error as defined above

Interpretation: If we repeated the experiment many times, $(1-\alpha)$% of intervals would contain the true $\mu_B - \mu_A$.

2.5 Sample Size Calculation

Theorem 2.2 (Required Sample Size per Group)**.** To detect effect size d with significance α and power $1-\beta$:

$$\boxed{n = 2 \cdot \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d}\right)^2} \tag{47}$$

Parameters:

- $z_p = \Phi^{-1}(p)$: p-quantile of standard normal $\mathcal{N}(0,1)$
- $z_{0.975} \approx 1.96$ (for $\alpha = 0.05$ two-tailed)
- $z_{0.80} \approx 0.84$ (for power = 80%)
- d: Expected effect size (Cohen's d)

Example 2.1 (Sample Size Calculation)**.** For $d = 0.5$ (medium effect), $\alpha = 0.05$, power $= 0.80$:

$$n = 2 \cdot \left(\frac{1.96 + 0.84}{0.5}\right)^2 \tag{48}$$
$$= 2 \cdot \left(\frac{2.80}{0.5}\right)^2 \tag{49}$$
$$= 2 \cdot (5.6)^2 \tag{50}$$
$$= 2 \cdot 31.36 \tag{51}$$
$$= 62.72 \approx \boxed{64 \text{ per group}} \tag{52}$$

3 Demo 3: Document Coherence Evaluation

Notation 3.1 (Document Structure)**.**

$$D = (s_1, s_2, \ldots, s_N) \qquad \text{Document with } N \text{ sentences} \tag{53}$$
$$W(s) = \{w : w \in s, w \notin \text{stopwords}, |w| > 2\} \qquad \text{Content words of sentence } s \tag{54}$$

3.1 Lexical Cohesion

Definition 3.1 (Word Frequency Across Document)**.** For word w:

$$\text{count}(w) = \sum_{i=1}^{N} \mathbf{1}[w \in s_i] \cdot \text{freq}(w, s_i) \tag{55}$$

where $\text{freq}(w, s_i)$ counts occurrences of w in sentence s_i.

Definition 3.2 (Lexical Cohesion Score)**.**

$$\boxed{\text{LC} = \frac{\displaystyle\sum_{w:\text{count}(w)>1} \text{count}(w)}{\displaystyle\sum_{w \in W_{\text{all}}} \text{count}(w)}} \tag{56}$$

Components:

- $W_{\text{all}} = \bigcup_{i=1}^{N} W(s_i)$: All content words in document
- Numerator: Total count of words appearing more than once (repeated words)
- Denominator: Total count of all content words

Range: $[0, 1]$ where 1 = all words are repeated, 0 = all words unique.

3.2 Semantic Coherence via Word Overlap

Definition 3.3 (Jaccard Similarity)**.** For two sets A and B:

$$\boxed{J(A, B) = \frac{|A \cap B|}{|A \cup B|}} \tag{57}$$

Properties:

- $J(A, B) \in [0, 1]$
- $J(A, B) = 1 \iff A = B$

- $J(A, B) = 0 \iff A \cap B = \emptyset$
- Symmetric: $J(A, B) = J(B, A)$

Definition 3.4 (Consecutive Sentence Overlap)**.**

$$\boxed{\text{overlap}(s_i, s_{i+1}) = J(W(s_i), W(s_{i+1})) = \frac{|W(s_i) \cap W(s_{i+1})|}{|W(s_i) \cup W(s_{i+1})|}} \tag{58}$$

Definition 3.5 (Semantic Coherence Score)**.**

$$\boxed{\text{SC} = \frac{1}{N-1} \sum_{i=1}^{N-1} \text{overlap}(s_i, s_{i+1})} \tag{59}$$

Intuition: Average word overlap between consecutive sentence pairs. High SC indicates smooth topic flow with shared terminology.

Example 3.1 (Coherent vs. Incoherent Document)**. Coherent:**

1. s_1: "The <u>bank</u> announced record profits today."
2. s_2: "The financial institution has been growing steadily."
3. s_3: "Some analysts remain skeptical about the <u>bank</u>."
4. s_4: "The <u>bank</u>'s CEO addressed these concerns."

$W(s_3) = \{\text{analysts, skeptical, bank}\}$, $W(s_4) = \{\text{bank, ceo, addressed, concerns}\}$

$$\text{overlap}(s_3, s_4) = \frac{|\{\text{bank}\}|}{|\{\text{analysts, skeptical, bank, ceo, addressed, concerns}\}|} = \frac{1}{6} \approx 0.167 \tag{60}$$

Incoherent:

1. "The bank announced profits." $\rightarrow W = \{\text{bank, announced, profits}\}$
2. "The river flowed past." $\rightarrow W = \{\text{river, flowed, past}\}$

$$\text{overlap}(s_1, s_2) = \frac{|\emptyset|}{|\{\text{bank, announced, profits, river, flowed, past}\}|} = \frac{0}{6} = 0 \tag{61}$$

3.3 Overall Coherence Score

Definition 3.6 (Weighted Combination)**.**

$$\boxed{\text{Coherence} = \sum_{m \in \mathcal{M}} w_m \cdot \text{score}_m} \tag{62}$$

Default weights:

Metric m	Weight w_m	Measures
Lexical Cohesion (LC)	0.25	Word repetition consistency
Semantic Coherence (SC)	0.35	Topic continuity
Entity Consistency (EC)	0.30	Same entity $\rightarrow$ same terms
Discourse Connectors (DC)	0.10	Logical flow markers

3.4 Why BLEU Fails for Discourse

Proposition 3.1 (BLEU Insensitivity to Pronouns)**.** Let:

$$f_p \approx 0.03 \qquad \text{Frequency of pronouns in typical text} \tag{63}$$

$$\Delta_{\text{acc}} = 0.5 \qquad \text{Accuracy improvement for pronoun translation} \tag{64}$$

Expected BLEU improvement:

$$\boxed{\Delta\text{BLEU} \approx f_p \cdot \Delta_{\text{acc}} = 0.03 \times 0.5 = 0.015} \tag{65}$$

This 1.5% change is within BLEU's noise range (±2%), making improvements invisible.

4 Demo 4: Attention Complexity Analysis

Notation 4.1 (Document and Model Parameters)**.**

$$N = \text{number of sentences in document} \tag{66}$$

$$L = \text{average sentence length (tokens)} \tag{67}$$

$$d = \text{model hidden dimension} \tag{68}$$

$$h = \text{number of attention heads} \tag{69}$$

$$d_k = d/h \quad \text{(dimension per head)} \tag{70}$$

$$k = \text{local context window (sentences)} \tag{71}$$

4.1 Self-Attention Mechanism

Definition 4.1 (Query, Key, Value Projections)**.** For input $\mathbf{X} \in \mathbb{R}^{T \times d}$ (sequence of T tokens):

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^Q \in \mathbb{R}^{T \times d_k} \qquad \text{(Query matrix)} \tag{72}$$

$$\mathbf{K} = \mathbf{X}\mathbf{W}^K \in \mathbb{R}^{T \times d_k} \qquad \text{(Key matrix)} \tag{73}$$

$$\mathbf{V} = \mathbf{X}\mathbf{W}^V \in \mathbb{R}^{T \times d_v} \qquad \text{(Value matrix)} \tag{74}$$

where $\mathbf{W}^Q, \mathbf{W}^K \in \mathbb{R}^{d \times d_k}$ and $\mathbf{W}^V \in \mathbb{R}^{d \times d_v}$.

Definition 4.2 (Scaled Dot-Product Attention)**.**

$$\boxed{\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}} \tag{75}$$

Step-by-step:

1. $\mathbf{Q}\mathbf{K}^\top \in \mathbb{R}^{T \times T}$: Pairwise similarity scores
2. Divide by $\sqrt{d_k}$: Scaling factor
3. softmax: Normalize each row to sum to 1
4. Multiply by $\mathbf{V}$: Weighted combination of values

Theorem 4.1 (Scaling Factor Derivation)**.** For random vectors $\mathbf{q}, \mathbf{k} \in \mathbb{R}^{d_k}$ with i.i.d. components $q_i, k_i \sim \mathcal{N}(0, 1)$:

$$\mathbf{q}^\top \mathbf{k} = \sum_{i=1}^{d_k} q_i k_i \tag{76}$$

$$\mathbb{E}[\mathbf{q}^\top \mathbf{k}] = \sum_{i=1}^{d_k} \mathbb{E}[q_i]\mathbb{E}[k_i] = 0 \tag{77}$$

$$\mathrm{Var}(\mathbf{q}^\top \mathbf{k}) = \sum_{i=1}^{d_k} \mathrm{Var}(q_i k_i) = \sum_{i=1}^{d_k} 1 = d_k \tag{78}$$

Scaling by $\sqrt{d_k}$ gives $\mathrm{Var}\left(\frac{\mathbf{q}^\top \mathbf{k}}{\sqrt{d_k}}\right) = 1$, preventing softmax saturation.

4.2 Full Concatenation Complexity

Theorem 4.2 (Full Attention Complexity)**.** For a document with N sentences of average length L, total sequence length is $T = NL$.

$$\boxed{\text{Time Complexity:}} \quad O(T^2 \cdot d) = O(N^2 L^2 d) \tag{79}$$

$$\boxed{\text{Memory Complexity:}} \quad O(T^2) = O(N^2 L^2) \quad \text{(attention weights)} \tag{80}$$

Derivation of Time Complexity.

1. $\mathbf{QK}^\top$: $(T \times d_k) \times (d_k \times T) = O(T^2 d_k)$ operations
2. Softmax: $O(T^2)$ (exponentiate and normalize each row)
3. Attention $\times \mathbf{V}$: $(T \times T) \times (T \times d_v) = O(T^2 d_v)$ operations

Total: $O(T^2 d_k + T^2 + T^2 d_v) = O(T^2 d) = O(N^2 L^2 d)$ □

4.3 Hierarchical Attention Complexity

Definition 4.3 (Two-Level Hierarchy)**.**

1. **Local Attention:** Each sentence attends to k previous sentences
2. **Global Attention:** Sentence embeddings attend to each other

Theorem 4.3 (Hierarchical Complexity)**.**

$$\text{Local:} \quad O(N \cdot (kL)^2 \cdot d) = O(Nk^2 L^2 d) \tag{81}$$

$$\text{Global:} \quad O(N^2 \cdot d) \tag{82}$$

$$\boxed{\text{Total:}} \quad O(Nk^2 L^2 d + N^2 d) \tag{83}$$

For small k (e.g., $k = 3$): $O(NL^2 d + N^2 d)$

Proof. **Local attention per sentence:**

- Context length: $(k+1) \cdot L$ tokens (current + k previous sentences)
- Attention complexity: $O(((k+1)L)^2 d) = O(k^2 L^2 d)$
- For N sentences: $O(Nk^2 L^2 d)$

Global attention:

- N sentence embeddings, each of dimension d
- Full attention: $O(N^2 d)$

□

4.4 Speedup Analysis

Theorem 4.4 (Theoretical Speedup)**.**

$$\boxed{\text{Speedup} = \frac{O(N^2L^2d)}{O(Nk^2L^2d + N^2d)} = \frac{N^2L^2}{Nk^2L^2 + N^2} = \frac{NL^2}{k^2L^2 + N}} \tag{84}$$

Example 4.1 (Numerical Speedup)**.** For $N = 50$, $L = 30$, $k = 3$, $d = 512$:

Full concatenation:

$$(NL)^2 \cdot d = (50 \times 30)^2 \times 512 = 1500^2 \times 512 = 2,250,000 \times 512 \approx 1.15 \times 10^9 \tag{85}$$

Hierarchical:

$$\text{Local:} \quad N \cdot k^2 \cdot L^2 \cdot d = 50 \times 9 \times 900 \times 512 = 207,360,000 \tag{86}$$

$$\text{Global:} \quad N^2 \cdot d = 2500 \times 512 = 1,280,000 \tag{87}$$

$$\text{Total:} \quad \approx 208,640,000 \tag{88}$$

$$\text{Speedup} = \frac{1.15 \times 10^9}{2.09 \times 10^8} \approx \boxed{5.5\times} \tag{89}$$

4.5 Memory Analysis

Theorem 4.5 (Memory Requirements)**.** For attention weights stored as float32 (4 bytes):

$$\text{Full:} \quad (NL)^2 \times 4 \text{ bytes} \tag{90}$$

$$\text{Hierarchical:} \quad ((k+1)L)^2 \times 4 + N^2 \times 4 \text{ bytes} \tag{91}$$

Example 4.2 (Memory Comparison)**.** $N = 50$, $L = 30$, $k = 3$:

$$\text{Full:} \quad 1500^2 \times 4 = 9,000,000 \text{ bytes} = 8.6 \text{ MB} \tag{92}$$

$$\text{Hierarchical:} \quad 120^2 \times 4 + 2500 \times 4 = 57,600 + 10,000 = 67,600 \text{ bytes} \approx 0.06 \text{ MB} \tag{93}$$

$$\text{Memory Reduction:} \approx \boxed{130\times} \tag{94}$$

5 Demo 5: LLM Document Processing

5.1 Autoregressive Language Modeling

Definition 5.1 (Conditional Probability Decomposition)**.** For sequence $Y = (y_1, y_2, \ldots, y_T)$:

$$\boxed{P(Y) = \prod_{t=1}^{T} P(y_t \mid y_1, y_2, \ldots, y_{t-1}) = \prod_{t=1}^{T} P(y_t \mid y_{<t})} \tag{95}$$

This is the **chain rule of probability** applied to sequences.

Definition 5.2 (Conditional Generation)**.** Given input document D, generate output $\hat{Y}$:

$$\boxed{P(\hat{Y} \mid D; \theta) = \prod_{t=1}^{m} P(\hat{y}_t \mid \hat{y}_{<t}, D; \theta)} \tag{96}$$

where θ are model parameters.

5.2 Summarization

Definition 5.3 (Maximum Likelihood Decoding)**.**

$$\hat{Y}^* = \text{argmax}_{\hat{Y}}\, P(\hat{Y} \mid D) = \text{argmax}_{\hat{Y}} \sum_{t=1}^{m} \log P(\hat{y}_t \mid \hat{y}_{<t}, D) \tag{97}$$

Definition 5.4 (Beam Search)**.** Maintain top-B partial hypotheses at each step:

$$\mathcal{H}_t = \text{top-}B\,\{(h, y) : h \in \mathcal{H}_{t-1}, y \in \mathcal{V}, \text{score}(h) + \log P(y \mid h, D)\} \tag{98}$$

where $\text{score}(h) = \sum_{i=1}^{|h|} \log P(h_i \mid h_{<i}, D)$.

5.3 Question Answering

Definition 5.5 (Extractive QA)**.** Find answer span (i, j) where $1 \le i \le j \le |D|$:

$$\boxed{(i^*, j^*) = \text{argmax}_{i \le j}\, P_{\text{start}}(i \mid D, Q) \cdot P_{\text{end}}(j \mid D, Q)} \tag{99}$$

where start/end probabilities are computed via softmax over token positions.

Definition 5.6 (Generative QA)**.** Generate answer A conditioned on document and question:

$$P(A \mid D, Q) = \prod_{t=1}^{|A|} P(a_t \mid a_{<t}, D, Q; \theta) \tag{100}$$

6 Demo 6: LLM-as-Judge Evaluation

6.1 Multi-Dimensional Scoring

Definition 6.1 (Evaluation Dimensions)**.**

$$\mathcal{D} = \{\text{quality}, \text{completeness}, \text{conciseness}, \text{coherence}, \text{factuality}\} \tag{101}$$

Definition 6.2 (Raw Score)**.** For each dimension $d \in \mathcal{D}$:

$$\text{score}_d(\hat{Y} \mid X) \in \{1, 2, 3, 4, 5\} \tag{102}$$

where X is source document and $\hat{Y}$ is generated text.

Definition 6.3 (Normalized Score)**.**

$$\boxed{\tilde{s}_d = \frac{\text{score}_d - 1}{4} \in [0, 1]} \tag{103}$$

Maps $\{1, 2, 3, 4, 5\} \to \{0, 0.25, 0.5, 0.75, 1\}$.

Definition 6.4 (Overall Score)**.**

$$\boxed{\text{Overall} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \tilde{s}_d} \tag{104}$$

6.2 Pairwise Comparison

Definition 6.5 (Preference Function)**.** Given source X and responses A, B:

$$\text{Pref}(A, B \mid X) \in \{A, B, \text{TIE}\} \tag{105}$$

with confidence $\text{conf} \in \{\text{LOW}, \text{MEDIUM}, \text{HIGH}\}$.

Definition 6.6 (Bradley-Terry Model)**.** Assume each response has latent quality $\pi_A, \pi_B > 0$:

$$\boxed{P(A \succ B) = \frac{\pi_A}{\pi_A + \pi_B} = \frac{1}{1 + \pi_B/\pi_A}} \tag{106}$$

Definition 6.7 (Elo Rating System)**.** **Expected score** for player A against B:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} \tag{107}$$

Update rule after A wins:

$$R'_A = R_A + K(1 - E_A) \qquad \text{(A's rating increases)} \tag{108}$$
$$R'_B = R_B + K(0 - E_B) \qquad \text{(B's rating decreases)} \tag{109}$$

where K is the update factor (typically $K = 32$).

Proposition 6.1 (Elo-Bradley-Terry Connection)**.** The Elo system is equivalent to Bradley-Terry with:

$$\pi_i = 10^{R_i/400} \tag{110}$$

Proof.

$$P(A \succ B) = \frac{\pi_A}{\pi_A + \pi_B} = \frac{10^{R_A/400}}{10^{R_A/400} + 10^{R_B/400}} = \frac{1}{1 + 10^{(R_B - R_A)/400}} = E_A \tag{111}$$

□

6.3 Tournament Ranking

Definition 6.8 (Round-Robin Tournament)**.** For n responses, conduct all $\binom{n}{2} = \frac{n(n-1)}{2}$ pairwise comparisons.

Definition 6.9 (Win Count)**.**

$$W_i = \sum_{j \neq i} [\mathbf{1}[\text{Pref}(i, j) = i] + 0.5 \cdot \mathbf{1}[\text{Pref}(i, j) = \text{TIE}]] \tag{112}$$

Definition 6.10 (Copeland Ranking)**.**

$$\text{rank}(i) = 1 + \sum_{j \neq i} \mathbf{1}[W_j > W_i] \tag{113}$$

7 Demo 7: API Architecture

7.1 REST API Design

Definition 7.1 (Endpoint Specification)**.**

Method	Endpoint	Function
GET	`/health`	Health check
POST	`/api/v1/summarize`	f : Document $\to$ Summary
POST	`/api/v1/enhance`	f : Text $\times$ Aspects $\to$ EnhancedText
POST	`/api/v1/ask`	f : Document $\times$ Question $\to$ Answer
POST	`/api/v1/evaluate`	f : Generated $\times$ Reference $\to$ Scores
POST	`/api/v1/judge`	f : Source $\times$ Summary $\to$ Evaluation
POST	`/api/v1/judge/compare`	f : Source $\times$ A $\times$ B $\to$ Preference

8 Advanced Topics

8.1 Hierarchical Context Modeling

Definition 8.1 (Local Context Attention)**.** For sentence s_t, local context is k previous sentences:

$$C_{\text{local}} = [s_{t-k}, s_{t-k+1}, \ldots, s_{t-1}] \tag{114}$$

Cross-attention:

$$\boxed{\alpha_{ij} = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j^c / \sqrt{d_k})}{\sum_\ell \exp(\mathbf{q}_i^\top \mathbf{k}_\ell^c / \sqrt{d_k})}, \quad \mathbf{c}_{\text{local}} = \sum_j \alpha_{ij} \mathbf{v}_j^c} \tag{115}$$

Definition 8.2 (Global Context Attention)**.** Sentence embedding via mean pooling:

$$\mathbf{s}_i = \frac{1}{|s_i|} \sum_{t=1}^{|s_i|} \mathbf{h}_t^{(i)} \tag{116}$$

Global attention:

$$\boxed{\beta_j = \text{softmax}(\mathbf{q}_{\text{sent}}^\top \mathbf{s}_j), \quad \mathbf{c}_{\text{global}} = \sum_{j=1}^{N} \beta_j \mathbf{s}_j} \tag{117}$$

Definition 8.3 (Gated Fusion)**.**

$$\boxed{g = \sigma(W_l \mathbf{c}_{\text{local}} + W_g \mathbf{c}_{\text{global}} + W_m \mathbf{h} + b)} \tag{118}$$

$$\boxed{\mathbf{c}_{\text{final}} = g \odot \mathbf{c}_{\text{local}} + (1 - g) \odot \mathbf{c}_{\text{global}}} \tag{119}$$

Symbol explanation:

- $g \in (0, 1)^d$: Gate vector (learned weighting)
- $\odot$: Element-wise (Hadamard) product
- $\sigma(x) = 1/(1 + e^{-x})$: Squashes to $(0, 1)$

8.2 Contrastive Evaluation

Definition 8.4 (Contrastive Pair)**.**
- Source: x (sentence with ambiguous pronoun)
- Context: c (contains antecedent)
- Correct: y^+ (proper pronoun translation)
- Contrastive: $\{y_1^-, \ldots, y_k^-\}$ (incorrect alternatives)

Definition 8.5 (Log-Probability Score)**.**

$$\boxed{\text{score}(y \mid x, c) = \sum_{t=1}^{|y|} \log P(y_t \mid y_{<t}, x, c; \theta)} \tag{120}$$

Definition 8.6 (Contrastive Accuracy)**.**

$$\boxed{\text{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1} \left[\text{score}(y_i^+) > \max_j \text{score}(y_{ij}^-) \right]} \tag{121}$$

8.3 Zero-Resource Domain Adaptation

Definition 8.7 (Implicit Domain Embedding)**.** From document context:

$$\boxed{\mathbf{d}_{\text{implicit}} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{h}_t^c} \tag{122}$$

Definition 8.8 (FiLM Conditioning)**.** Feature-wise Linear Modulation:

$$\boxed{\mathbf{h}' = \gamma(\mathbf{d}) \odot \mathbf{h} + \beta(\mathbf{d})} \tag{123}$$

where $\gamma, \beta : \mathbb{R}^{d_{\text{domain}}} \to \mathbb{R}^d$ are learned MLPs.

Initialization: $\gamma \to \mathbf{1}$, $\beta \to \mathbf{0}$ (identity transformation at start).

8.4 Oracle Experiments

Oracle experiments establish **upper bounds** on potential improvements from discourse modeling.

Definition 8.9 (Oracle Upper Bound Theorem)**.** If providing oracle signal $\mathcal{O}$ improves BLEU by Δ, then perfect modeling of the corresponding phenomenon can improve BLEU by **at most** Δ:

$$\boxed{\Delta_{\text{perfect}} \leq \Delta_{\text{oracle}} = \text{BLEU}_{\text{oracle}} - \text{BLEU}_{\text{baseline}}} \tag{124}$$

Definition 8.10 (Oracle Types)**.** Three oracle signals for discourse phenomena:

1. **Previous Target Oracle:** Gold standard previous translation

$$\text{Input} = \text{prev_target} \oplus \text{source} \tag{125}$$

2. **Coreference (Pronoun) Oracle:** Perfect gender/number from target

$$\text{Input} = \text{XPRONOUN} + \text{target_pronoun} \oplus \text{source} \tag{126}$$

3. **Coherence Oracle:** Perfect word sense disambiguation

$$\text{Input} = \text{target_translation}(w_{\text{repeated}}) \oplus \text{source} \tag{127}$$

where $\oplus$ denotes concatenation with separator.

Definition 8.11 (Impact Analysis)**.**

$$\Delta_{\text{abs}} = \text{BLEU}_{\text{oracle}} - \text{BLEU}_{\text{baseline}} \qquad \text{(Absolute gain)} \tag{128}$$

$$\Delta_{\text{rel}} = \frac{\Delta_{\text{abs}}}{\text{BLEU}_{\text{baseline}}} \times 100\% \qquad \text{(Relative gain)} \tag{129}$$

Example 8.1 (WMT Results (Stojanovski & Fraser, 2018))**.**

Oracle Type	Baseline	Oracle	Δ BLEU
Coreference (pronoun)	28.57	35.59	**+7.02**
Coherence (repeated words)	28.57	30.46	+1.89
Previous target sentence	28.57	30.35	+1.78

Key insight: Coreference has the highest potential (+7 BLEU), indicating models significantly under-utilize gender/number information from context.

Remark 8.1 (Fair vs. Noisy Oracles)**.**

- **Fair Oracle:** Uses only source-side information (realistic but weaker signal)
- **Noisy Oracle:** Adds extra target information (tests model's ability to extract relevant signal from noise)

9 Summary of Key Formulas

Concept	**Formula**
ROUGE-L F_1	$F = \dfrac{(1+\beta^2)PR}{R+\beta^2 P}$ where $P = \ell/m$, $R = \ell/n$
BLEU	$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_n w_n \log p_n\right)$
Welch's t	$t = \dfrac{\bar{Y} - \bar{X}}{\sqrt{s_A^2/n_A + s_B^2/n_B}}$
Cohen's d	$d = \dfrac{\bar{Y} - \bar{X}}{\sqrt{(s_A^2 + s_B^2)/2}}$
Sample Size	$n = 2\left(\dfrac{z_{1-\alpha/2} + z_{1-\beta}}{d}\right)^2$
Jaccard Similarity	$J(A,B) = \dfrac{\lvert A \cap B \rvert}{\lvert A \cup B \rvert}$
Full Attention	$O(N^2 L^2 d)$
Hierarchical Attention	$O(Nk^2 L^2 d + N^2 d)$
Scaled Dot-Product	$\text{Attn}(Q,K,V) = \text{softmax}\left(\dfrac{QK^\top}{\sqrt{d_k}}\right)V$
Gated Fusion	$\mathbf{c} = g \odot \mathbf{c}_{\text{local}} + (1-g) \odot \mathbf{c}_{\text{global}}$
Bradley-Terry	$P(A \succ B) = \dfrac{\pi_A}{\pi_A + \pi_B}$
Oracle Upper Bound	$\Delta_{\text{perfect}} \leq \text{BLEU}_{\text{oracle}} - \text{BLEU}_{\text{baseline}}$
Contrastive Accuracy	$\text{Acc} = \frac{1}{N}\sum_i \mathbf{1}[\text{score}(y^+) > \max_j \text{score}(y_j^-)]$